

Creating Hash Values

Project description

HASH

The following code demonstrates how I used Linux commands

- Compare the plain text of the two files presented for hashing
- Compute the sha256sum hash of the two separate files
- Compare the hashes provided to identify the differences

I used the "ls" command to list the contents of the directory.

```
analyst@ece21falce3f:~$ ls
file1.txt  file2.txt
analyst@ece21falce3f:~$ cat file1.txt
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
analyst@ece21falce3f:~$ cat file2.txt
X5O!P%@AP[4\PZX54(P^)7CC)7}$EICAR-STANDARD-ANTIVIRUS-TEST-FILE!$H+H*
analyst@ece21falce3f:~$
```

Two files, "file1.txt" and "file2.txt", are listed. Use the "cat" command to display the contents of the "file1.txt" file

I Used the "sha256sum" command to generate the hash of the "file1.txt & 2

```
analyst@ece21falce3f:~$ sha256sum file1.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267  file1.txt
analyst@ece21falce3f:~$ sha256sum file2.txt
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
analyst@ece21falce3f:~$
```

Next i Used the "sha256sum" command to generate the hash of the "file1.txt" file, and send the output to a new file called "file1hash"

```
analyst@ece21falce3f:~$ sha256sum file1.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbdfd8267  file1.txt
analyst@ece21falce3f:~$ sha256sum file2.txt
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
analyst@ece21falce3f:~$ sha256sum file1.txt >> file1hash
analyst@ece21falce3f:~$ sha256sum file2.txt >> file2hash
analyst@ece21falce3f:~$ cmp file1hash file2hash
file1hash file2hash differ: char 1, line 1
analyst@ece21falce3f:~$
```

I Used the "cmp" command to highlight the differences in the "file1hash" and "file2hash"

```
analyst@ece21falce3f:~$ sha256sum file1.txt
131f95c51cc819465fa1797f6ccacf9d494aaaff46fa3eac73ae63ffbfdfd8267  file1.txt
analyst@ece21falce3f:~$ sha256sum file2.txt
2558ba9a4cad1e69804ce03aa2a029526179a91a5e38cb723320e83af9ca017b  file2.txt
analyst@ece21falce3f:~$ sha256sum file1.txt >> file1hash
analyst@ece21falce3f:~$ sha256sum file2.txt >> file2hash
analyst@ece21falce3f:~$ cmp file1hash file2hash
file1hash file2hash differ: char 1, line 1
analyst@ece21falce3f:~$
```

Summary

During this task, I successfully demonstrated proficiency in utilizing various command-line tools. Specifically, I utilized the `ls` command to list the contents of a directory, the `cmp` command to compare two hash values and determine their similarity, and the "`cat` command" to display the contents of a file.

By using the "`ls` command", I efficiently obtained a comprehensive list of the files within the directory, specifically identifying the presence of "`file1.txt`" and "`file2.txt`". This command allowed me to gain a quick overview of the directory's contents. Furthermore, I effectively employed the `cmp` command to compare two hash values.

This comparison allowed me to ascertain whether the two files possessed identical or differing contents, providing valuable insights into their integrity and potential discrepancies. Lastly, I employed the "`cat` command" to display the contents of "`file1.txt`".

The output allowed for the easy visualization of the file's contents, providing an efficient means of reviewing and examining the data within. This exercise not only showcased my familiarity with essential command-line tools but also demonstrated my ability to apply them in a professional and efficient manner.

--Connor Gibson